

Q.1) Draw a typical section of canal showing all components in detail. Also define them.

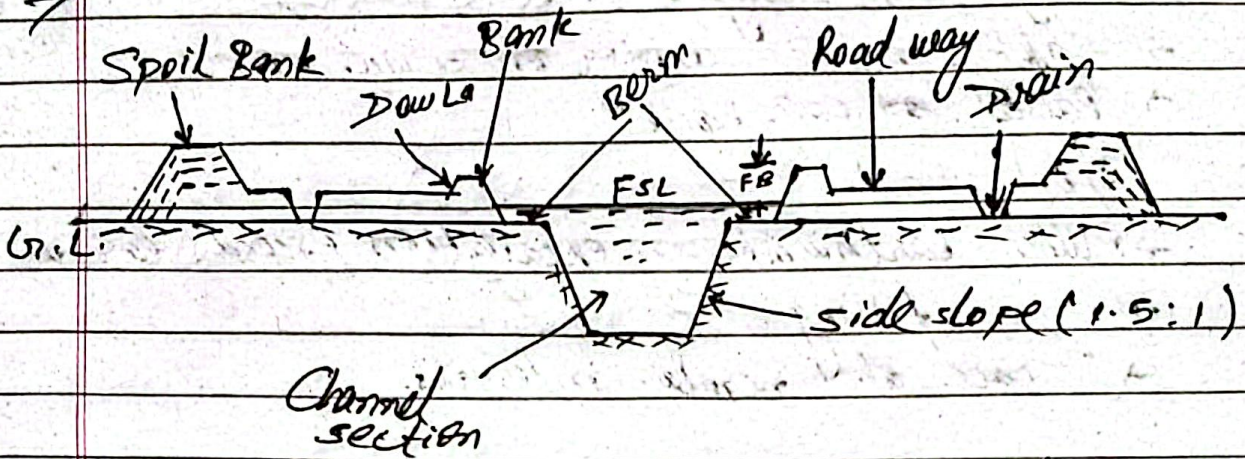


fig:- Typical section of canal.

1) Side slopes:

→ slope provided at channel section to make it efficient, stable & safe.

→ depends upon type of soil,

→ steeper slope is cutting.

→ 1.5:1 in cutting & 2:1 to 4:5 1.5:1 in filling.

2) Berms: Horizontal distant distance left at ground level betn toe of bank & edge of cutting.

→ Protects bank from erosion & breach.

→ Scope for future expansion

3) Freeboard: Margin betn full supply level (FSL) & bank level.

→ Depends upon canal size

4) Canal banks: - To retain water, provided obstructs at the edge of channel alongside.

Dowlas/Dowel:

- Small embankment by the side of service road
- Height generally 0.3m.
- Safety in driving on service roads
- Raised over bank.

Spoil bank:

- When earthwork in excavation exceeds ~~excess~~ in filling.
- space at regular intervals.

Q.2) How can you align canals? State with their merits & demerits.

⇒ Canal alignment means selecting the most suitable route or path for a canal so that irrigation water can be conveyed economically & efficiently.

The major methods are:

1. Ridge canal:

A canal is aligned along the ridge or watershed line.

Merits:

- Commands land on both sides.
- No cross-drainage works are generally required.
- Suitable for irrigation of large areas.

Demerits:

- Large initial cutting may be required.
- Difficult to maintain on uneven terrain.
- Not suitable where the ridge is highly irregular.

2. Contour Canal

A canal is aligned approximately along the contour line of the ground.

Merits:

- Requires less cutting & filling.
- Suitable for hilly & undulating areas.
- Can irrigate land on the lower side.

Demerits:

- Requires many cross-drainage works.
- Risk of water logging on the lower side.
- Alignment may become longer.

3. Side slope canal

A canal is aligned across the contour, generally along the side slope of a valley.

Merits:

- Suitable for hilly & mountainous regions.
- Comparatively fewer cross-drainage works than contour canals.
- Can follow a relatively direct route.

Demerits:

- Requires considerable cutting & filling.
- Difficult construction & maintenance.
- Risk of slope instability.

Q.3) Determine the average boundary shear stress for a hydraulically efficient trapezoidal canal carrying a discharge of 15 cumecs with side slopes 1.5H:1V & bed of 0.3° lined with concrete.

⇒

Given; Channel lining = concrete.

$$Q = 15 \text{ m}^3/\text{s}$$

$$m = 1.5 \text{ \& } S = \tan 0.3^\circ = 0.005236.$$

For most hydraulically efficient trapezoidal canal,

$$\text{Hydraulic radius (R)} = y/2$$

$$\text{wetted area (A)} = (B + my) \times y = (B + 1.5y)y$$

$$\begin{aligned} \therefore \text{perimeter (P)} &= B + 2y\sqrt{1+m^2} \\ &= B + 2y\sqrt{1+1.5^2} \\ &= B + 3.6055y. \end{aligned}$$

for efficient channel,

The top width $T = B + 2my$
& condition is

$$\frac{T}{2} = y\sqrt{1+m^2}$$

$$\frac{B + 2my}{2} = y\sqrt{1+m^2}$$

$$B \approx 0.6055y$$

$$\begin{aligned} \therefore A &= (0.6055y + 1.5y)y \\ &= 2.1055y^2 \end{aligned}$$

$$\begin{aligned} P &= 0.6055y + 3.6055y \\ &= 4.211y \end{aligned}$$

So,

$$\frac{A}{P} = R \Rightarrow \frac{2.1055y^2}{4.211y} = \frac{y}{2}$$

3) Manning's roughness coeff.

for concrete lining $n = 0.012 \text{ to } 0.015$

so let's take $n = 0.014$.

$$Q = \frac{1}{n} A R^{2/3} S^{1/2}$$

$$15 = \frac{1}{0.014} \cdot (2.1055y^2) \cdot \left(\frac{y}{2}\right)^{2/3} \sqrt{0.005236}$$

$$y = 1.341 \text{ m.}$$

$$\therefore R = y/2 = 0.6705 \text{ m.}$$

So,

Average boundary shear stress (τ_0)
(on channel bed)

$$\begin{aligned} \tau_0 &= \gamma_w R S \\ &= 9810 \times 0.6705 \times 0.005236 \\ &= 34.44 \text{ N/m}^2. \end{aligned}$$

Q.4) Design an irrigation channel to carry 100 cumecs of discharge for bed slope of 1 in 2500. Adopt critical velocity ratio of 1.05 & Kutter's rugosity coefficient of 0.024.

⇒ Required expressions;

$$V_0 = 0.55 y^{0.64} \times m \quad \text{--- (1) } (\because \text{Kennedy's formula})$$

m is C.V.R = 1.05.

$$A = Q/V_0 = \frac{100}{V_0} \quad \text{--- (2)}$$

$$A = By + my^2 \text{ \& find } B. \quad \text{--- (3)}$$

$$P = B + 2y\sqrt{1+m^2} \quad \text{--- (4)}$$

$$R = A/P. \quad \text{--- (5)}$$

Kutter's formula;

$$V = \left[\frac{\frac{1}{N} + \left(23 + \frac{0.00155}{S}\right)}{1 + \left(23 + \frac{0.00155}{S}\right) \frac{N}{\sqrt{R}}} \right] \sqrt{RS}$$

from given data;

$$V = \left[\frac{\frac{1}{0.024} + 23 + \frac{0.00155}{1/2500}}{1 + \left(23 + \frac{0.00155}{1/2500}\right) \frac{0.024}{\sqrt{R}}} \right] \times \sqrt{\frac{R}{2500}}$$

$$\therefore V = \left[\frac{68.541}{1 + \frac{0.645}{\sqrt{R}}} \right] \times \sqrt{\frac{R}{2500}} \quad \text{--- (6)}$$

If $V_0 < V$; increase y .

$V_0 > V$; decrease y .

&

repeat the process untill $V_0 \approx V$.

So,

Iter.	Assumed y (m)	V_0 (m/s) ①	A (m ²) ②	B (m) ③	P (m) ④	R (m) ⑤	V (m/s) ⑥	Cond ⁿ	Remark
1.	1	0.5775	178.16	172.16	174.988	0.989	0.827	$V_0 < V$	$y \uparrow$
2.	2	0.899	111.19	53.59	59.211	1.876	1.276	$V_0 < V$	$y \uparrow$
3.	3	1.166	85.72	25.57	34.05	2.516	1.546	$V_0 < V$	$y \uparrow$
4.	4	1.402	71.30	18.82	25.14	2.836	1.669	$V_0 < V$	$y \uparrow$
5.	5	1.617	61.816	7.363	21.505	2.874	1.683	$V_0 < V$	$y \uparrow$
6.	5.5	1.719	58.158	5.074	20.63	2.819	1.662	$V_0 > V$	$y \downarrow$
7.	5.3	1.679	59.534	5.936	20.921	2.845	1.672	$V_0 > V$	$y \downarrow$
8.	5.275	1.674	59.734	6.049	20.96	2.848	1.674	$V_0 = V$	y

The depth of canal (y) = 5.275 for 1:1 side slope.

Q.5) Derive the relation $S = \left[\frac{f^{5/3}}{3340 Q^{1/6}} \right]$ for Lacey's regime theory.

⇒ Fundamental Lacey eqns.

1) Velocity formula $V = \sqrt{\frac{2}{5} f R}$

Squaring both sides:

$$V^2 = \frac{2}{5} f R \Rightarrow R = \frac{5V^2}{2f} \quad \text{--- (1)}$$

2. Wetted Perimeter formula:

$$P = 4.75 \sqrt{R} \quad \text{--- (2)}$$

3) Regime slope formula:

$$S = \frac{f^{5/3}}{3340 R^{1/6}} \quad \text{--- (3)}$$

Lacey's general Regime equation is

$$V = 10.8 R^{2/3} S^{1/3}$$

we have $V = \sqrt{\frac{2}{5} f R} \quad \text{--- (1)}$

$$V = \left(\frac{Q f^2}{140} \right)^{1/6} \quad \text{--- (ii)}$$

Equating (1) & (ii)

$$\sqrt{\frac{2}{5} f R} = \left(\frac{Q f^2}{140} \right)^{1/6}$$

$$R^{1/2} = \frac{2}{5} \left(\frac{5}{2f} \right)^{1/2} \left(\frac{Q f^2}{140} \right)^{1/6} \quad \text{--- (iii)}$$

substituting the value of $R^{1/2}$ in $S = \frac{0.0002 f^{3/2}}{R^{1/2}}$

$$S = \frac{0.0002 f^{3/2}}{\left(\frac{5}{2f} \right)^{1/2} \times \left(\frac{Q f^2}{140} \right)^{1/6}}$$

$$S = \frac{0.0002 f^{3/2}}{\frac{\sqrt{5}}{\sqrt{2} \times f^{1/2}} \times \frac{Q^{1/6} \times f^{1/3}}{140^{1/6}}}$$

$$S = \frac{0.0002 \times 140^{1/6}}{\left(\frac{\sqrt{5}}{\sqrt{2}} \right)} \times \frac{f^{3/2 + 1/2 - 1/3}}{Q^{1/6}}$$

$$S = \frac{f^{5/3}}{3340 Q^{1/6}} \quad \text{proved.}$$

Next,

$$S = \frac{0.0002 f^{3/2}}{R^{1/2}} \quad \text{is from,}$$

$$V = 10.8 R^{2/3} S^{1/3} \text{ \& } V = \sqrt{\frac{2}{5} f R}$$

Q.6) Design an irrigation channel to carry 40 cumecs of discharge with $B/D = 2.5$ (i.e., width/depth). The critical velocity ratio is 1. Assume a suitable value of Kutter's roughness coefficient & use Kennedy's method.

→ let us take kutter's rugity coeff. is 0.024.

$$\therefore N = 0.024$$

$$m = 1 \quad (\because \text{CVR})$$

$$B/D = 2.5$$

$$\text{side slope} = 1H:2V; \quad z = \frac{1}{2} = 0.5$$

$$\text{Now, Area (A)} = by + zy^2 BD + zD^2$$

$$= D^2(B/D + 0.5)$$

By Kennedy's formula;

$$V_0 = 0.55 m D^{0.64}$$

$$= 0.55 D^{0.64}$$

$$\therefore Q = AV_0 = D^2(2.5 + 0.5) \times 0.55 D^{0.64}$$

$$Q = 1.65 D^{2.64}$$

$$\therefore 40 = 1.65 D^{2.64}$$

$$D = 3.345 \text{ m.}$$

$$\therefore \text{Bed width (B)} = 2.5 \times 3.345$$

$$= 8.363 \text{ m}$$

So,

$$\text{Area (A)} = 3.345^2 \times 3 = 33.567 \text{ m}^2$$

$$P = B + 2D\sqrt{1+m^2} = 8.363 + 2 \times 3.345\sqrt{1+1}$$

$$= 17.824 \text{ m.}$$

$$R = \frac{A}{P} = \frac{33.567}{17.824} = 1.883 \text{ m} \cdot 2.118 \text{ m}$$

By Kennedy's formula;

$$V = 0.55 \times 1 \times 3.345^{0.64}$$

$$= 1.1911 \text{ m/s.}$$

Now, using kutter's formula;

$$V = \left\{ \frac{\frac{1}{N} + 23 + \frac{0.00155}{S}}{1 + \left(23 + \frac{0.00155}{S} \right) \frac{N}{\sqrt{R}}} \right\} \times \sqrt{RS}$$

$$1.1911 = \left\{ \frac{\frac{1}{0.024} + 23 + \frac{0.00155}{S}}{1 + \left(23 + \frac{0.00155}{S} \right) \frac{0.024}{\sqrt{2.118}}} \right\} \times \sqrt{2.118 \times S}$$

for 1st trial.

$$\therefore S = 3.047 \times 10^{-4}$$

$$= 1:3282$$

or,

If we don't neglect these term;

$$S = 1:3396.$$

Q.7) Design a regime channel for a discharge of 75 cumecs & silt factor 1.08, using Lacey's theory.

$$\Rightarrow Q = 75 \text{ m}^3/\text{s} \quad d_{\text{mm}} = \text{particle size,}$$

$$f = 1.08 \quad (\because \text{if not given; } f = 1.76\sqrt{d_{\text{mm}}})$$

$$S.S = 1:2 \quad (H:V) \quad \text{i.e., } z = 0.5.$$

$$\text{velocity of flow (V)} = \left(\frac{Q f^2}{140} \right)^{1/6} = \left(\frac{75 \times 1.08^2}{140} \right)^{1/6}$$

$$= 0.925 \text{ m/s}$$

$$R = \frac{5}{2} \left(\frac{V^2}{f} \right) = \frac{5}{2} \times \frac{0.925^2}{1.08}$$

$$= 1.979 \text{ m.}$$

$$\text{wetted perimeter (P)} = 4.75\sqrt{Q}$$

$$= 4.75 \times \sqrt{75}$$

$$= 41.136 \text{ m.}$$

$$\text{Bed slope (S)} = \frac{f^{5/3}}{3340 \times Q^{1/6}} = \frac{1.08^{5/3}}{3340 \times 75^{1/6}}$$

$$= 1.657 \times 10^{-4}$$

$$= 1:6033$$

$$\text{Area (A)} = \frac{Q}{V} = \frac{50}{0.869} = 81.081 \text{ m}^2.$$

$$81.081 = by + 0.5y^2 \quad \text{--- (1)}$$

$$P = b + 2y\sqrt{1+m^2}$$

$$41.1363 = b + 2y \times \sqrt{1+0.5^2}$$

$$b = 41.136 - y\sqrt{5}$$

put b on (1).

$$81.081 = (41.136 - \sqrt{5}y)y + 0.5y^2$$

$$\boxed{y = 2.17 \text{ m.}}$$

$$\therefore b = 41.136 - 2.17\sqrt{5}$$

$$\boxed{b = 36.284 \text{ m}}$$

Q.7) Design a regime channel for a discharge of 75 cumecs. Curves & si.

Q.8) Water flows at a depth of 0.6 m in a wide stream having a bed slope of 1 in 2500. the median diameter of the sand bed is 1 mm. Determine whether the soil grains are stationary or moving & comment as to whether the stream bed is scouring or non-scouring. (hint: use mittal & swamee formula).

$$\rightarrow \text{depth of flow (y)} = 0.6 \text{ m}$$

$$\rightarrow \text{Bed slope (S)} = \frac{1}{2500} = 0.0004.$$

$$\rightarrow \text{Median diameter of sand bed (d)} = 1 \text{ mm} = 0.001 \text{ m.}$$

Now,

=> Avg. Boundary shear stress (τ_0):

For a wide stream, the hydraulic radius (R) is approximately equal to the depth of flow (y = 0.6 m).

$$\therefore \tau_0 = \rho_w \cdot R \cdot S$$

$$\text{using the } \rho_w = 9810 \text{ N/m}^3.$$

$$\tau_0 = 9810 \times 0.6 \times 0.0004 = 2.3544 \text{ N/m}^2.$$

Again Critical shear stress (τ_c) using mittal & Swamee formula;

$$\tau_c = 0.155 + \frac{0.409 d_{mm}^2}{\sqrt{1 + 0.177 d_{mm}}}$$

$$\tau_c = 0.155 + \frac{0.409 \cdot d^{2.58}}{\sqrt{1 + 0.177 d^{2.58}}} = 0.532$$

where d is in mm.

$$\therefore \tau_c = 0.155 + 0.409 \times 1^{0.56} = 0.532 \text{ N/m}^2$$

Here, $\tau_o > \tau_c$

i.e. the hydrodynamic shear force applied by the flowing water exceeds the critical force required to move the sediment particles.

So the soil grains are moving & the stream bed is scouring.

Q.9) An Agr. irrigation channel is to be constructed in coarse alluvium gravel with avg. grain size of 5 cm. The channel has to carry 3 cumecs of discharge & the longitudinal slope is 0.01. The banks of the channel will be protected by grass against scouring. Find the minimum width of channel. (hint: use Strickler's & Manning's formula).

*)

$$\text{Discharge } (Q) = 3 \text{ m}^3/\text{s}$$

$$\text{Longitudinal slope } (s) = 0.01$$

$$\text{avg. grain size } (d) = 5 \text{ cm} = 0.05 \text{ m}$$

Bed is protected by grass against scouring

tt Finding the Manning's roughness coefficient (n) using Strickler's formula;

$$n = \frac{d^{1/6}}{24.0}$$

$$n = \frac{(0.05)^{1/6}}{24.0} = 0.02876 \approx 0.0253$$

Now,

Critical shear stress for stable non-scouring channel;

$$\begin{aligned} \tau_c &= 0.056 \rho_w d (s-1) \\ &= 0.056 \times 9810 \times 0.05 \times (2.65-1) \\ &= 45.322 \text{ N/m}^2 \end{aligned}$$

Now we know, avg. shear stress

$$\tau_o = \rho_w R S \leq \tau_c$$

$$R = \frac{\tau_c}{\rho_w S} = \frac{45.322}{9810 \times 0.01}$$

$$R = 0.462 \text{ m}$$

Now using velocity;

$$V = \frac{1}{n} R^{2/3} s^{1/2}$$

$$V = \frac{1}{0.0253} \times 0.462^{2/3} \times 0.01^{1/2}$$

$$V = 2.362 \text{ m/s}$$

Now, area (A) = $\frac{Q}{V} = \frac{8}{2.362}$
 $= 1.27 \text{ m}^2$.

For a wide rectangular or efficient trapezoidal channel.

$B = \frac{A}{R}$ ($\because R \approx y$ on wide section)

$= \frac{1.27}{0.462} = 2.75 \text{ m}$.

Q.10) (Refer SK Chong example 4.3 for solution):
 A canal is to be designed to carry a discharge of 5.6 cumec. The slope of canal is 1 in 1000. The soil is coarse alluvium having a grain size of 5mm. Assuming the canal to be inclined & of trapezoidal section, determine a suitable section for canal. Assume $\phi = 37^\circ$ & $\theta = 30^\circ$. (Hint: use formula for unprotected side slopes case II & calculate discharge using manning's eqn with trial & error for dimensions).

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Q.4) An unlined canal giving seepage loss of 3.3 cumec per million sq meters of wetted area is proposed to be lined with 10cm thick cement concrete lining, which costs Rs 180 per 10 sqm. The life of lining is 40 years. Given the following data, work out the economics of lining & benefit cost ratio.

Annual revenue per cumec of water from all crops = Rs. 3.5 lakhs.

Discharge in the channel = 83.5 cumec.

wetted perimeter of channel = 18.8 m.

wetted perimeter of the lining = 18.5 m.

Annual maintenance cost of unlined channel per 10 sqm = Rs. 1.
(Assume, 40% of maintenance cost is saved in lining, seepage loss after lining is 0.01 cumecs per million sqm of wetted area & interest rate @ 5% p.a).

Assume any suitable data if required.

(Hint: Perform calculations considering 1 km length of canal.)

⇒ Length of canal (L) = 1000 m

wetted perimeter of unlined channel (P_u) = 18.8 m.

wetted perimeter of lined channel (P_l) = 18.5 m.

Unlined seepage loss rate = 3.3 cumec per mill. m^2 .

Lined seepage loss rate = 0.01 cumec per mill. m^2 .

Value of water (R_1) = Rs. 3.5 lakhs/cumec

Maintenance cost of unlined channel = Rs 1 / $10m^2$.

Maintenance cost saved (p) = 40% = 0.40

Cost of lining = Rs 180 / $10m^2$.

Life of lining (Y) = 40 years

Interest rate (r) = 5% p.a.

Now

⇒ Wetted Area of unlined canal (A_1) = $P_1 \times L = 18.8 \times 1000$
= $18,800 m^2$.

Wetted Area of lined canal (A_2) = $P_2 \times L = 18.5 \times 1000$
= $18,500 m^2$

⇒ Water saved by lining (m);

$$= \frac{3.3 \times 18,800 - 0.01 \times 18,500}{1000000}$$

$$m = 0.061855 \text{ cumec.}$$

⇒ Total Annual benefit (AB)
using formula;
 $AB = mR_1 + pR_2$

1. revenue from water saved ($m.R_1$)

$$m.R_1 = 0.061855 \times 3,50,000$$
$$= Rs. 21,649.25$$

2. Annual maintenance cost of unlined canal (R_2)

$$R_2 = \frac{18800}{10} \times 1 = Rs. 1,880$$

3. Maintenance cost saved ($p.R_2$)

$$p.R_2 = 0.40 \times 1880 = Rs. 752.$$

4. Total Annual Benefit (AB):

$$AB = 21,649.25 + 752 = Rs. 22,401.25$$

⇒ Total Annual Cost (AC)
using the formula;

$$AC = \frac{C}{Y} + 0.5.C \cdot \frac{r}{100}$$

1. Capital expenditure for lining (C):

$$C = \frac{18,500}{10} \times 180 = Rs. 3,33,000$$

2. Annual depreciation charge (C/Y):

$$\text{Depreciation} = \frac{3,33,000}{40} = Rs. 8,325.$$

3. Interest on capital expenditure

$$\frac{0.5.C \cdot r}{100} = 0.5 \times 3,33,000 \times \frac{5}{100} = Rs. 8,325$$

4. Total annual cost (AC):

$$AC = 8325 + 8325 = \text{Rs. } 16650$$

$$\Rightarrow \text{Benefit cost ratio} = \frac{AB}{AC}$$

$$= \frac{22401.25}{16650}$$

$\approx 1.345 > 1$ so the proposed canal lining is economically justified.

Q.12 Design a lined channel to carry a discharge of 15 cumecs. The available & accepted country slope is 1 in 9000. Assume suitable values of side slopes & good brick work in lining.

$\Rightarrow$ Let us take Manning's roughness coeff. (n) = 0.015 for brick lining

Let us assume the channel section is most efficient trapezoidal due to small discharge.

And side slope be 1.25H:1V.

$$S = 1:9000$$

$$\tan \theta = \frac{1}{1.25}$$

$$\therefore \cot \theta = 1.25$$

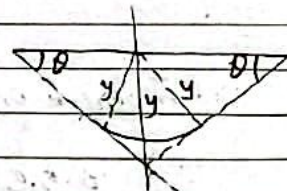


fig: 1.

$$\theta = 0.675 \text{ rad.}$$

$$A = y^2 (\theta + \cot \theta) = (0.675 + 1.25) y^2 = 1.925 y^2$$

$$P = 2y (\theta + \cot \theta) = (0.675 + 1.25) 2y = 3.85 y$$

$$R = \frac{y}{2} = 0.5 y$$

Now using Manning's formula:

$$Q = \frac{1}{n} \cdot A \cdot R^{2/3} \cdot S^{1/2}$$

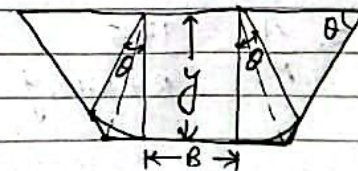
$$15 = \frac{1}{0.015} \cdot (1.925 y^2) (0.5 y)^{2/3} \cdot \frac{1}{\sqrt{9000}}$$

$$y = 2.93 \text{ m.}$$

Hence the section shown in fig: 1 with 2.93m depth & 1.25:1 side slope.

Q.13) Design a concrete lined trapezoidal channel to carry a discharge of 350 cumecs at a slope of 1 in 5000. The side slopes of the channel may be taken as 1.5:1, the value of n for lining is 0.014. Assume limiting velocity in the channel as 2 m/s. (Ans: $y = 3.10 \text{ m}$, $B = 49.8 \text{ m}$)

$\Rightarrow$



$$\Rightarrow V = \frac{1}{n} \cdot R^{2/3} \cdot S^{1/2}$$

$$2 = \frac{1}{0.014} R^{2/3} \cdot \frac{1}{\sqrt{5000}}$$

$$R = 2.79 \text{ m.}$$

The channel section is assumed to be trapezoidal as shown in fig for 1.5 : 1 slopes.

$$\text{So, } \cot \theta = 1.5 \text{ \& } \theta = 33.7^\circ = 0.588 \text{ radians}$$

$$\text{Using } A = y(B + y\theta + y \cot \theta) = By + y^2(\theta + \cot \theta)$$

$$P = B + 2y\theta + 2y \cot \theta = B + 2y(\theta + \cot \theta)$$

$$\therefore A = y(B + 0.59y + 1.5y)$$

$$= y(B + 2.09y)$$

$$P = B + 1.18y + 3y = B + 4.18y$$

$$\text{But, } A = \frac{Q}{V} = \frac{350}{2} = 175 \text{ sq.m.}$$

$$\therefore 175 = y(B + 2.09y)$$

$$B = \frac{175}{y} - 2.09y$$

Now,

$$R = \frac{A}{P} = \frac{175}{B + 4.18y}$$

$$2.79 = \frac{175}{\left(\frac{175 - 2.09y}{y}\right) + 4.18y}$$

$$\therefore y = 3.2 \text{ m.}$$

$$\text{But, } B = \frac{175}{y} - 2.09y$$

$$= \frac{175}{3.2} - 2.09 \times 3.2$$

$$= 48 \text{ m.}$$

$$\therefore \boxed{y = 3.2}$$

$$\boxed{B = 48 \text{ m}}$$